

Statistics :

Statistics deals with the methods for collection, classification and analysis of numerical data for drawing valid conclusions and making reasonable decisions.

Variable (or) Variate :

A quantity which can vary from one individual to another is called a variable or variate.

Ex: heights, weights, ages, wages of persons, rainfall records of cities etc.

Continuous Variable:

Quantities which can take any numerical value within a certain range are called Continuous variables.

Ex: as the child grows, his/her height takes all possible values from 50 cm to 100 cm.

Discrete Variable (or) discontinuous variable:

Quantities which are incapable of taking all possible values are called discrete or

discontinuous variables.

Ex:

The number of children of a man can have positive integers 1, 2, 3, ... etc. (no value between any two consecutive integers)

Collection of Data:

The collection of data constitutes the starting point of any statistical investigation.

Data may be collected for each & every unit of the whole lot (population), for it would ensure greater accuracy.

But Complete enumeration is prohibitively expensive and time consuming.

As such out of a very large number of items, a few of them (a sample) are selected and conclusions drawn on the basis of this sample are taken to hold for the population.

Raw data (or) Ungrouped data

(or) primary data (or) Source data

(or) atomic data :

Raw data is data that has not been processed for use.

Classification of data:

The data collected in the course of an inquiry is not in an easily assimilable form. As such, its proper classification is necessary for making intelligent inferences.

The classification is done by dividing the raw data into a convenient number of groups (classes) according to the values of the variable and find the frequency of the variable in each group.

Ex:

Consider the raw data relating to marks obtained in mathematics by 64 students.

79	88	75	60	93	71	59	85
84	75	82	68	90	62	88	76
65	75	87	74	62	95	78	63
78	82	75	91	77	69	74	68
67	73	81	72	63	76	75	85
80	73	57	88	78	62	76	53
62	67	97	78	85	76	65	71
78	89	61	75	95	60	79	83

This data can conveniently be grouped and shown in a tabular form as follows.

Class	Frequency	Cumulative frequency
50-54	1	1
55-59	2	3
60-64	9	12
65-69	7	19
70-74	8	27
75-79	17	44
80-84	6	50
85-89	8	58
90-94	3	61
95-99	3	64
	<u>64</u>	

From the above table we can see that there is one student getting marks between 50-54, two students getting marks between 55-59, nine students getting marks between 60-64 and so on.

Thus the 64 figures, have been put into only 10 groups, called the classes. The width of the class is called the class interval and the number in that interval is called the frequency. The mid-point or mid-value of the class is called the class mark.

The above table showing the classes and the corresponding frequencies is called a frequency table.

Thus a set of raw data summarised by distributing it into a number of classes along with their frequencies is known as a frequency distribution.

While forming a frequency distribution the number of classes should not ordinarily exceed 20, and should not, in general, be less than 10. As far as possible, the class intervals should be of equal width.

Cumulative frequency:

In some investigations, we require the number of items less than a certain value.

We add up the frequencies of the classes up to that value and call this number as the cumulative frequency. In the above table, the third column shows the cumulative

frequencies, i.e. the number of students getting less than 54 marks, less than 59 marks and so on.

Exclusive and Inclusive class intervals:

Class-intervals of the type $\{x : a \leq x < b\} = [a, b)$ are called exclusive since they exclude the upper limit of the class. The following data are classified on this basis.

Income (£)	No. of persons
50-100	88
100-150	70
150-200	52
200-250	30
250-300	23

In this method, the upper limit of one class is the lower limit of the next class. In this example, there are 88 persons whose income is from ₹ 50 to ₹ 99.99. A person whose income is ₹ 100 is included in the class ₹ 100 - ₹ 150.

Class intervals of type $\{x : a \leq x \leq b\}$
 $= [a, b]$ are called inclusive
since they include the upper
limit of the class.

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Class intervals of type $\{x : a \leq x \leq b\}$
 $= [a, b]$ are called inclusive
since they include the upper
limit of the class.

The following data are
classified on this basis.

Income (₹)	No. of persons
50-99	60
100-149	38
150-199	22
200-249	16
250-299	7

However, to ensure continuity and to get correct class-limits, exclusive method of classification should be adopted. To convert inclusive class-intervals into exclusive we have to make an adjustment.

Adjustment.

Find the difference between the lower limit of the second class and the upper limit of the

first class. Divide it by 2.

Subtract the value so obtained from all the lower limits and add the value to all upper limits.

In the above example, the adjustment factor is $\frac{100-99}{2} = 0.5$

The adjusted classes would then be as follows:

Income (£)	No. of persons
49.5 - 99.5	60
99.5 - 149.5	38
149.5 - 199.5	22
199.5 - 249.5	16
249.5 - 299.5	7

The size of the class interval is 50.

Comparison of frequency distributions.

The Condensation of data in the form of a frequency distribution is very useful as far as it brings a long series of observations into a compact form. But in practice, we are generally interested in comparing two or more series.

The inherent inability of the human mind to grasp in its entirety even the data in the form of a frequency distribution compels us to seek for certain constants which could concisely give an insight into the important characteristics of the series.

The chief constants which summarise the fundamental characteristics of the frequency distributions are

- (i) Measures of central tendency
- (ii) Measures of dispersion
- (iii) Measures of skewness.

Averages (or) Measure of central
Tendency (or) measure of location.

According to professor Bowley, averages are "Statistical Constants which enable us to Comprehend in a single effort the significance of the whole". They give us an idea about the Concentration of the values in the central part of the distribution.

plainly speaking, an average of a statistical series is the value of the variable which is the representative of the entire distribution. The following are the five measures of central tendency that are in common use:

- 1) Arithmetic mean (or) Simply mean
- 2) Median
- 3) Mode
- 4) Geometric mean
- 5) Harmonic mean

Measure of central tendency :

A frequency distribution in general, shows clustering of the data around some central value. Finding of this central value or the average is of importance, as it gives a most representative value of the whole group. Different methods give different averages which are known as the measures of central tendency. The commonly used measures of central value are Mean, median, Mode, Geometric mean and Harmonic mean.

Mean:

1. If $x_1, x_2, \dots, x_n$ are a set of n values of a variate, then the arithmetic mean (or simply mean) is given by

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{\sum x_i}{n}$$

2. In Case of frequency distribution,

$$\text{mean} = \frac{x_1 f_1 + x_2 f_2 + \dots + x_n f_n}{N}$$

$$= \frac{\sum f_i x_i}{N}, \quad N = \sum f_i$$

Note:

In case of continuous frequency distribution, x_i is taken as the mid-value of the corresponding class

① Find Arithmetic mean of the following data.

1, 3, 5, 7, 8.

Sol:

$$\begin{aligned}\text{Mean } \bar{x} &= \frac{\sum x_i}{n} \\ &= \frac{1+3+5+7+8}{5} \\ &= \frac{24}{5} = 4.8\end{aligned}$$

② Find Arithmetic mean of the following frequency distribution.

$x :$	1	2	3	4	5	6	7
$f :$	5	9	12	17	14	10	6

Sol:

$$\text{Mean } \bar{x} = \frac{\sum f_i x_i}{N}$$

x	f	xf
1	5	5
2	9	18
3	12	36
4	17	68
5	14	70
6	10	60
7	6	42
	<u>73</u>	<u>299</u>

$$\begin{aligned}\bar{x} &= \frac{\sum f_i x_i}{N} \\ &= \frac{299}{73} \\ &= 4.096.\end{aligned}$$

③ Calculate Arithmetic mean of the following table.

Marks:	0-10	10-20	20-30	30-40	40-50	50-60
No. of students:	12	18	27	20	17	6

Sol:

Marks x	No. of Students f	xf	
5	12	60	$\bar{x} = \frac{\sum f_i x_i}{N}$
15	18	270	
25	27	675	$= \frac{2800}{100}$
35	20	700	
45	17	765	$= 28.$
55	6	330	
	<u>100</u>	<u>2800</u>	

Short cut method:

$$\bar{x} = A + \frac{\sum f_i d_i}{N}$$

where A is an arbitrary origin
and $d_i = x_i - A$.

Ex:

x	f	$d = x - A$ $d = x - 3$	fd
1	5	-2	-10
2	9	-1	-9
3	12	0	0
4	17	1	17
5	14	2	28
6	10	3	30
7	6	4	24
	<u>73</u>		<u>80</u>

$$\begin{aligned}\bar{x} &= A + \frac{\sum f_i d_i}{N} \\ &= 3 + \frac{80}{73} \\ &= 3 + 1.096 \\ &= 4.096,\end{aligned}$$

Step-deviation method:

$$\bar{x} = A + h \frac{\sum f_i u_i}{N}$$

Where $u_i = \frac{x_i - A}{h}$

A is an arbitrary origin
h is the magnitude of the class interval

Ex:

x	f	$u = \frac{x-A}{h} = \frac{x-25}{10}$	fu
5	12	-2	-24
15	18	-1	-18
25	27	0	0
35	20	1	20
45	17	2	34
55	6	3	18
	<u>100</u>		<u>30</u>

$$\bar{x} = A + h \left(\frac{\sum fu}{N} \right)$$

$$= 25 + 10 \left(\frac{30}{100} \right)$$

$$= 25 + 3$$

$$= 28.$$

Mean of the Composite Series:

If $\bar{x}_i$ ($i=1, 2, \dots, k$) be the arithmetic means of k distributions with respective frequencies n_i ($i=1, 2, \dots, k$) then mean $\bar{x}$ of the whole distribution obtained by combining the k distributions is given by

$$\bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2 + \dots + n_k\bar{x}_k}{n_1 + n_2 + \dots + n_k}$$

- ① The mean annual salary paid to all employees of a company was £ 5000. The mean annual salaries paid to male and female employees were £ 5200 and £ 4200 respectively. Determine the Percentage

of males and females employed
by the company.

Sol:

Let n_1 and n_2 represent
percentage of males and females
respectively.

$$\text{Then } n_1 + n_2 = 100 \rightarrow \textcircled{1}$$

Mean annual salary of all employees
 $(\bar{x}) = ₹ 5000$

Mean annual salary of all males
 $(\bar{x}_1) = ₹ 5200$

Mean annual salary of all females
 $(\bar{x}_2) = ₹ 4200$

$$\bar{x} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}$$

$$\text{ie } 5000 = \frac{5200n_1 + 4200n_2}{100}$$

$$52n_1 + 42n_2 = 5000$$

$$26n_1 + 21n_2 = 2500 \rightarrow \textcircled{2}$$

$$\textcircled{2} \Rightarrow 26n_1 + 21n_2 = 2500$$

$$\textcircled{1} \times 26 \Rightarrow \begin{array}{r} 26n_1 + 26n_2 = 2600 \\ (-) \quad \quad \quad (-) \quad \quad \quad (-) \\ \hline -5n_2 = -100 \end{array}$$

$$\boxed{n_2 = 20}$$

$$\therefore n_1 = 100 - n_2$$

$$= 100 - 20$$

$$\boxed{n_1 = 80}$$

Hence the percentage of males and females is 80 and 20 respectively.

Merits and Demerits of Arithmetic

mean:

Merits:

- 1) It is rigidly defined.
- 2) It is easy to understand and easy to calculate.
- 3) It is based upon all the observations.
- 4) It is amenable to algebraic treatment.
- 5) Of all the averages, arithmetic mean is affected least by fluctuations of sampling. This property is sometimes described by saying

that arithmetic mean is a stable average.

Demerits

1) It cannot be determined by inspection nor it can be located graphically.

2) Arithmetic mean cannot be used if we are dealing with qualitative characteristics which cannot be measured quantitatively; such as intelligence, honesty, beauty, etc.

In such cases median is the only average to be used.

3) Arithmetic mean Cannot be obtained if a single observation is missing or lost or is illegible unless we drop it out and compute the arithmetic mean of the remaining values.

4) Arithmetic mean is affected very much by extreme values. In case of extreme items, arithmetic mean gives a distorted picture of the distribution and no longer remains representative of the distribution.

5) Arithmetic mean may lead to wrong conclusions if the details of the data from which it is computed are not given.

6) Arithmetic mean Cannot be Calculated

if the extreme class is open. Ex:
below 10 or above 90. Moreover, even
if a single observation is missing
mean Cannot be calculated.

7) In extremely asymmetrical (skewed)
distribution, usually arithmetic mean
is not a suitable measure of location.

Median:

Median of a distribution is the value of the variable which divides it into two equal parts.

In Case of ungrouped data, if the number of observations is odd then median is the middle value after the values have been arranged in ascending (or) descending order of magnitude.

In Case of even number of observations, there are two middle terms and median is obtained by taking the arithmetic mean of the middle terms.

① Find the median of 25, 20, 15, 35,

Ascending order 15, 20, 25, 35

$n = 5$ (odd)

median = $\left(\frac{n+1}{2}\right)^{\text{th}}$ item median is 20.
 $= 3^{\text{rd}}$ item = 20

② Find the median of 8, 20, 50, 25,
15, 30.

$n = 6$

$\frac{n}{2} = \frac{6}{2} = 3^{\text{rd}}$ item Ascending order 8, 15, 20, 25, 30, 50

$\frac{n}{2} + 1 = 4^{\text{th}}$ item = 25

median = $\frac{20+25}{2} = 22.5$

In Case of discrete frequency distribution, median is obtained by considering Cumulative frequency. The steps for calculating median are given below.

1. Find $\frac{N}{2}$, $N = \sum f_i$

2. See the cumulative frequency just greater than $\frac{N}{2}$.

3. Corresponding value of x is median.

① Find the median for the following table.

x	1	2	3	4	5	6	7	8	9
f	8	10	11	16	20	25	15	9	6
cf	8	18	29	45	65	90	105	114	120

$$N = 120$$

$$\frac{N}{2} = \frac{120}{2} = 60$$

Cumulative frequency just greater than $\frac{N}{2}$ is 65.

The value of x corresponding to 65 is 5.
 $\therefore$ median is 5.

Median for Continuous frequency distribution:

In the case of Continuous frequency distribution, the class corresponding to the cumulative frequency just greater than $N/2$ is called the median class and the value of median is obtained by the following formula.

$$\text{Median} = l + \frac{h}{f} (N/2 - c)$$

Where l is the lower limit of median class
 f is the frequency of the median class
 h is the magnitude of the median class

'C' is the cumulative frequency of the class preceding the median class, and $N = \sum f$.

① Find the median wage of the following distribution.

wages (in Rs)	2000-3000	3000-4000	4000-5000
No. of workers	3	5	20
	5000-6000	6000-7000	
	10	5	

Sol.

wages (in Rs)	No. of workers	C.f.
2000-3000	3	3
3000-4000	5	8
4000-5000	20	28
5000-6000	10	38
6000-7000	5	43
	<u>43</u>	

$$\text{Here } \frac{N}{2} = \frac{43}{2} = 21.5$$

Cumulative frequency just greater than 21.5 is 28 and the corresponding class is 4000-5000.

Thus median class is 4000-5000

$$\text{median} = l + \frac{h}{f} \left(\frac{N}{2} - c \right)$$

$$l = 4000$$

$$f = 20$$

$$h = 1000$$

$$c = 8$$

$$= 4000 + \frac{1000}{20} (21.5 - 8)$$

$$= 4000 + 675$$

$$= 4675.$$

- ② In a factory employing 3000 persons, 5 percent earn less than Rs. 3 per hour, 580 earn from Rs. 3.0 to Rs. 4.50 per hour, 30 percent earn from Rs. 4.51 to Rs. 6.00 per hour, 500 earn from Rs. 6.01 to Rs. 7.50

per hour, 20 percent earn from Rs. 7.51 to Rs. 9.00 per hour, and the rest earn Rs. 9.01 or more per hour. What is the median wage?

Sol:

The given information can be expressed in tabular form as follows.

Earnings (in Rs)	percentage of workers	No. of workers f	c.f	class boundaries $\frac{3.01-3}{2} = 0.005$
less than 3	5%	$\frac{5}{100} \times 3000 = 150$	150	Below 3.005
3.01-4.50	—	580	730	3.005-4.505
4.51-6.00	30%	$\frac{30}{100} \times 3000 = 900$	1630	4.505-6.005
6.01-7.50	—	500	2130	6.005-7.505
7.51-9.00	20%	$\frac{20}{100} \times 3000 = 600$	2730	7.505-9.005
9.01 or more	—	$3000 - 2730 = 270$	3000	9.005 or more

$$N = \sum f = 3000$$

$$\frac{N}{2} = \frac{3000}{2} = 1500$$

Cumulative frequency just greater than 1500 is 1630 and the corresponding class 4.51-6.00, whose class boundaries are 4.505-6.005, is the median class.

$$\text{Median} = l + \frac{h}{f} (N/2 - c)$$

$$= 4.505 + \frac{1.5}{900} (1500 - 730)$$

$$= 4.505 + 1.283$$

$$= 5.79.$$

Merits and Demerits of median:

Merits:

- 1) It is rigidly defined.
- 2) It is easily understood and is easy to calculate. In some cases it can

be located merely by inspection.

3) It is not at all affected by extreme values.

4) It can be calculated for distributions with open-end classes.

Demerits:

1) In case of even number of observations median cannot be determined exactly. we merely estimate it by taking the mean of two middle terms.

2) It is not based on all the observations. For example, the median of 10, 25, 50, 60 and 65 is 50. we can replace the observations 10 and 25 by any two values which are smaller than 50 and the observations 60 and 65 by any

two values which are greater than 50 without affecting the value of median. This property is sometimes described by saying that median is insensitive.

3) It is not amenable to algebraic treatment.

4) As compared with mean, it is affected much by fluctuations of sampling.

Uses:

1. Median is the only average to be used while dealing with qualitative data which cannot be measured quantitatively but still can be arranged in ascending or descending order of magnitude, example, to find the

average intelligence or average honesty among a group of people.

2. It is to be used for determining the typical value in problems concerning wages, distribution of wealth, etc.

Mode:

Let us consider the following statements:

(i) The average height of an Indian (male) is 5'-6".

(ii) The average size of the shoes sold in a shop is 7.

(iii) An average student in a hostel spends Rs. 150 p.m.

In all the above cases, the average referred to is mode.

Mode is the value which occurs most frequently in a set of observations and around which the other items of the set cluster densely.

In other words, mode is the value of the variable which is predominant in the series.

① Find the mode of 5, 6, 7, 6, 3, 2, 1.
mode is 6.

② Find the mode of 3, 2, 5, 6, 7, 3, 9, 6
mode are 3 & 6.

Note:

Two mode are called bimodal.
More than two mode are called multimodal.

In the case of discrete frequency distribution, mode is the value of x corresponding to maximum frequency.

① Find the mode for the following frequency distribution.

x :	1	2	3	4	5	6	7	8
f :	4	9	16	25	22	15	7	3

Sol:

Here maximum frequency is 25
and corresponding value of x is 4
 $\therefore$ mode is 4.

Note:

But in any one (or more) of the following cases:

- i) if the maximum frequency is repeated
 - ii) if the maximum frequency occurs in the very beginning or at the end of the distribution and
 - iii) if there are irregularities in the distribution,
- then the value of mode is determined by the method of grouping which is illustrated below by an example.

Example:

Find the mode of the following frequency distribution:

Size (x): 1 2 3 4 5 6 7 8 9 10 11 12

Frequency (f): 3 8 15 23 35 40 32 28 20 45 14 6

Sol:

Here we see that the distribution is not regular since the frequencies

are increasing steadily up to 40 and then decrease but the frequency 45 after 20 does not seem to be consistent with the distribution. Here we cannot say that since maximum frequency is 45, mode is 10. Here we shall locate mode by the method of grouping as explained below.

Size x	Frequency					
	(i)	(ii)	(iii)	(iv)	(v)	(vi)
1	3	11	23	26	46	73
2	8					
3	15	38		58		
4	23					
5	35	75	72	107	100	
6	40					
7	32	60	48	80	93	79
8	28					
9	20	65	59	65		
10	45					
11	14	20				
12	6					

The frequencies in Column (i) are the original frequencies. Column (ii) is obtained by combining the frequencies two by two. If we leave the first frequency and combine the remaining frequencies two by two we get column (iii). Combining the frequencies two by two after leaving the first two frequencies results in a repetition of Column (ii). Hence, we proceed to combine the frequencies three by three, thus getting Column (iv). The combination of frequencies three by three after leaving the first frequency results in column (v) and after leaving the first two frequencies results in column (vi).

To find mode we form the following table:

Analysis table

Column No.	Maximum frequency	Value or Combination of values of x giving maximum frequency in 2.
1	2	3
(i)	45	10
(ii)	75	5, 6
(iii)	72	6, 7
(iv)	98	4, 5, 6
(v)	107	5, 6, 7
(vi)	100	6, 7, 8

On examining the values in column 3 above, we find that the value 6 is repeated the maximum number of times and hence the value of mode is 6

and not 10 which is an irregular item.

In case of continuous frequency distribution, mode is given by the

formula:
$$\text{mode} = l + \frac{h(f_1 - f_0)}{(f_1 - f_0) - (f_2 - f_1)}$$

(or)

$$\text{mode} = l + \frac{h(f_1 - f_0)}{2f_1 - f_0 - f_2}$$

where l is the lower limit of the modal class, h is the magnitude of the modal class, f_1 is the frequency of the modal class, and f_0 and f_2 are the frequencies of the classes preceding and succeeding the modal class respectively.

① Find the mode for the following distribution.

Class interval: 0-10 10-20 20-30 30-40 40-50 50-60

Frequency : 5 8 7 12 28 20

60-70 70-80
10 10

Sol:

Here maximum frequency is 28.

Thus the class 40-50 is the modal class.

$$l = 40, h = 10$$

$$f_1 = 28, f_0 = 12, f_2 = 20$$

$$\text{Mode} = 40 + \frac{10(28-12)}{2 \times 28 - 12 - 20}$$

$$= 40 + 6.666$$

$$= 46.67.$$

Note:

1. For a symmetrical distribution, mean, median and mode coincide.
2. If the distribution is moderately asymmetrical, the mean, median and mode obey the following empirical relationship.

$$\text{Mean} - \text{Median} = \frac{1}{3} (\text{Mean} - \text{Mode})$$

$$\Rightarrow \text{Mode} = 3 \text{Median} - 2 \text{Mean}$$

Merits and Demerits of Mode:

Merits:

- 1) Mode is readily comprehensible and easy to calculate. Like

median, mode can be located in some cases merely by inspection.

2) Mode is not at all affected by extreme values.

3) Mode can be conveniently located even if the frequency distribution has class-interval of unequal magnitude provided the modal class and the classes preceding and succeeding it are of the same magnitude.

Open-end classes also do not pose any problem in the location of mode.

Demerits:

1) Mode is ill-defined. It is not always possible to find a clearly

defined mode. In some cases, we may come across distributions with two modes. Such distributions are called bi-modal. If a distribution has more than two modes, then it is said to be multimodal.

- 2) It is not based upon all the observations.
- 3) It is not capable of further mathematical treatment.
- 4) As compared with mean, mode is affected to a greater extent by fluctuations of sampling.

Uses:

Mode is the average to be used to find the ideal size, example, in business forecasting, in the manufacture of ready-made garments, shoes, etc.

Geometric mean:

Geometric mean of a set of n observations is the n th root of their product. Thus the geometric mean G , of n observations x_i , $i=1, 2, \dots, n$ is

$$G = (x_1 x_2 \dots x_n)^{1/n}$$

Taking log of both sides,

$$\log G = \frac{1}{n} (\log x_1 + \log x_2 + \dots + \log x_n)$$

$$\log G = \frac{1}{n} \sum_{i=1}^n \log x_i$$

$$\therefore G = \text{Antilog} \left(\frac{1}{n} \sum_{i=1}^n \log x_i \right)$$

In case of frequency distribution,

geometric mean G is given

by $G = (x_1^{f_1} x_2^{f_2} \dots x_n^{f_n})^{1/N}$, $N = \sum f_i$

Taking log of both sides, we get

$$\log G = \frac{1}{N} (f_1 \log x_1 + f_2 \log x_2 + \dots + f_n \log x_n)$$

$$\log G = \frac{1}{N} \sum_{i=1}^n f_i \log x_i$$

$$G = \text{Antilog} \left(\frac{1}{N} \sum_{i=1}^n f_i \log x_i \right)$$

In the case of continuous frequency distribution, x is taken to be the value corresponding to the mid-point of the class-intervals.

Merits and Demerits of geometric

mean:

- 1) It is rigidly defined.
- 2) It is based upon all the observations.
- 3) It is suitable for further mathematical treatment.
- 4) It is not affected much by fluctuations of sampling.
- 5) It gives comparatively more weight to small items.

Note:

If n_1 and n_2 are the sizes, G_1 and G_2 the geometric means of two series respectively, the geometric mean G , of the combined series is given by

$$\log G = \frac{n_1 \log G_1 + n_2 \log G_2}{n_1 + n_2}$$

Demerits:

- 1) Because of its abstract mathematical character, geometric mean is not easy to understand and to calculate for a non-mathematics person.
- 2) If any one of the observation is zero, geometric mean becomes zero and if any one of the observations is negative, geometric mean becomes imaginary regardless of the magnitude of the other items.

Uses:

- 1) Geometric mean is used to find the rate of population growth and the rate of interest.

2) Geometric mean used in the construction of index numbers.

Problems:

1) Find the geometric mean of the following values 15, 12, 13, 19, 10.

Solution:

Given data: 15, 12, 13, 19, 10

x	$\log x$
15	1.1761
12	1.0792
13	1.1139
19	1.2788
10	1.0000
5.648	

$$n = 5$$

$$\sum \log x = 5.648$$

$$G.M = \text{Antilog} \left(\frac{\sum \log x}{n} \right)$$

$$= \text{Antilog} \left(\frac{5.648}{5} \right)$$

$$= \text{Antilog} (1.1296)$$

$$= 13.48$$

② Calculate the geometric mean of the following data:

x	12	13	14	15	16	17
f	5	4	4	3	2	1

Solution:

x	f	$\log x$	$f \log x$
12	5	1.0792	5.3960
13	4	1.1139	4.4556
14	4	1.1461	4.5844
15	3	1.1761	3.5283
16	2	1.2041	2.4092
17	1	1.2304	1.2304
	<u>19</u>		<u>21.6029</u>

$$N = 19$$

$$\Sigma f \log x = 21.6029$$

$$G.M = \text{Antilog of } \frac{\Sigma f \log x}{N}$$

$$= \text{Antilog of } \frac{21.6029}{19}$$

$$= \text{Antilog of } 1.1371$$

$$= 13.71$$

③ Find the geometric mean of the following data:

Marks :	0-10	10-20	20-30	30-40	40-50
No. of students :	4	8	10	6	7

Solution:

class	Mid x	$\log x$	f	$f \log x$
0-10	5	0.6990	4	2.7960
10-20	15	1.1761	8	9.4088
20-30	25	1.3969	10	13.9790
30-40	35	1.5441	6	9.2646
40-50	45	1.6532	7	11.5724
			<u>35</u>	<u>47.0208</u>

$$N = 35$$

$$\sum f \log x = 47.0208$$

$$G.M = \text{Antilog of } \frac{\sum f \log x}{N}$$

$$= \text{Antilog of } \frac{47.0208}{35}$$

$$= \text{Antilog of } 1.3435$$

$$= 22.06$$

Harmonic Mean:

Harmonic mean H , of n observation x_i ,
 $i=1, 2, \dots, n$ is given by

$$H = \frac{1}{\frac{1}{n} \sum_{i=1}^n \left(\frac{1}{x_i} \right)}$$

In Case of frequency distribution,

$$H = \frac{1}{\frac{1}{N} \sum_{i=1}^n \left(\frac{f_i}{x_i} \right)}, \quad N = \sum_{i=1}^n f_i.$$

Merits:

- 1) Harmonic mean is rigidly defined.
- 2) It is based upon all the observations.
- 3) It is suitable for further mathematical treatment.
- 4) Like geometric mean, it is not affected much by fluctuations of sampling.

5) It gives greater importance to small items and is useful only when small items have to be given a greater weightage.

Demerits:

Harmonic mean is not easily understood and is difficult to compute.

Problems:

① Find the harmonic mean of the following data 8, 9, 6, 11, 10, 5.

Solution:

$$H = \frac{1}{\frac{1}{6} \left(\frac{1}{8} + \frac{1}{9} + \frac{1}{6} + \frac{1}{11} + \frac{1}{10} + \frac{1}{5} \right)}$$

$$H = \frac{6}{0.7936} = 7.560$$

② Given the following frequency distribution of first year students of a particular college, calculate the harmonic mean.

Age (years)	13	14	15	16	17
No. of Student	2	5	13	7	3

Solution:

x	f	f/x
13	2	0.1538
14	5	0.3571
15	13	0.8667
16	7	0.4375
17	3	0.1765
	<u>30</u>	<u>1.9916</u>

$$N = \sum f = 30, \quad \sum (f/x) = 1.9916$$

$$H = \frac{N}{\sum (f/x)} = \frac{30}{1.9916} = 15.0631 \approx 15 \text{ years.}$$

3) From the following data calculate Harmonic mean.

class-interval	10-20	20-30	30-40	40-50	50-60
Frequency	30	75	70	135	220

Solution:

class interval	frequency f	midvalue x	f/x
10-20	30	15	2
20-30	75	25	3
30-40	70	35	2
40-50	135	45	3
50-60	220	55	4
	<u>530</u>		<u>14</u>

$$H = \frac{N}{\sum(f/x)} = \frac{530}{14} = 37.86.$$

Partition Values:

These are the values which divide the series into a number of equal parts.

The three points which divide the series into four equal parts are called quantiles. The first, second and third points are known as the first, second and third quantiles Q_1 , Q_2 and Q_3 respectively.

The nine points which divide the series into ten equal parts are called deciles.

The ninety-nine points which divide the series into hundred equal parts are called percentiles.

Note:

1. First quartile (or) lower quartile is denoted by Q_1 .
2. Third quartile (or) upper quartile is denoted by Q_3 .
3. Second quartile is denoted by Q_2 which is the median.
4. D_5 is the median.
5. P_{50} is the median.

Examples:

① Find the quartiles for the following data.

i) 5, 7, 4, 4, 6, 2, 8.

Ascending order: 2, 4, 4, 5, 6, 7, 8

$$\text{Quantile 1 } (Q_1) = 4$$

Quantile 2 (Q_2), which is also the median = 5.

$$\text{Quantile 3 } (Q_3) = 7$$

ii) 1, 3, 3, 4, 5, 6, 6, 7, 8, 8.

$$Q_2 = \frac{5+6}{2} = 5.5$$

$$Q_1 = 3$$

$$Q_3 = 7$$

iii) 4, 17, 7, 14, 18, 12, 3, 16, 10, 4, 4, 11

Ascending order:

3, 4, 4, 4, 7, 10, 11, 12, 14, 16, 17, 18

$$Q_2 = \frac{10+11}{2} = 10.5$$

$$Q_1 = \frac{4+4}{2} = 4$$

$$Q_3 = \frac{14+16}{2} = 15$$

2) Find the quartiles for the following data. 25, 18, 30, 8, 15, 5, 10, 35, 40, 45.

Ascending order:

5, 8, 10, 15, 18, 25, 30, 35, 40, 45.

$$Q_1 = \left(\frac{n+1}{4}\right)^{\text{th}} \text{ item} \quad Q_i = i\left(\frac{n+1}{4}\right)^{\text{th}} \text{ item}$$

$$= \left(\frac{10+1}{4} \right)^{\text{th}} \text{ item}$$

$$= (2.75)^{\text{th}} \text{ item}$$

$$= 2^{\text{nd}} \text{ item} + \frac{3}{4} (3^{\text{rd}} \text{ item} - 2^{\text{nd}} \text{ item})$$

$$= 8 + \frac{3}{4} (10 - 8)$$

$$= 8 + \frac{3}{4} (2)$$

$$= 8 + 1.5$$

$$\boxed{Q_1 = 9.5}$$

$$Q_3 = 3 \left(\frac{n+1}{4} \right)^{\text{th}} \text{ item}$$

$$= 3 \times (2.75)^{\text{th}} \text{ item}$$

$$= (8.25)^{\text{th}} \text{ item}$$

$$= 8^{\text{th}} \text{ item} + \frac{1}{4} (9^{\text{th}} \text{ item} - 8^{\text{th}} \text{ item})$$

$$= 35 + \frac{1}{4} (40 - 35)$$

$$= 35 + 1.25 = 36.25$$

$$\boxed{Q_3 = 36.25}$$

Quantiles, Deciles and Percentiles for
ungrouped data:

$$Q_i = \frac{i(n+1)}{4} \text{th observation}$$

$$Q_1 = \frac{n+1}{4} \text{th observation}$$

$$Q_2 = \frac{2(n+1)}{4} \text{th observation}$$

$$Q_3 = \frac{3(n+1)}{4} \text{th observation.}$$

$$D_j = \frac{j(n+1)}{10} \text{th observation.}$$

$$D_5 = \frac{5(n+1)}{10} \text{th observation.}$$

$$P_k = \frac{k(n+1)}{100} \text{th observation}$$

$$P_{50} = \frac{50(n+1)}{100} \text{th observation.}$$

① Eight coins were tossed together and the number of heads resulting was noted. The operation was repeated 256 times and the frequencies (f) that were obtained for different values of x , the number of heads, are shown in the following table. Calculate median, quantiles, 4th decile and 27th percentile.

x :	0	1	2	3	4	5	6	7	8
f :	1	9	26	59	72	52	29	7	1

Solution:

x :	0	1	2	3	4	5	6	7	8
f :	1	9	26	59	72	52	29	7	1
cf :	1	10	36	95	167	219	248	255	256

$$\text{Here } \frac{N}{2} = \frac{256}{2} = 128$$

Cumulative frequency just greater than 128 is 167.

Thus, median = 4. i.e. $Q_2 = 4$

$$\text{Here } \frac{N}{4} = \frac{256}{4} = 64$$

Cumulative frequency just greater than 64 is 95.

Hence $Q_1 = 3$.

$$\text{Here } \frac{3N}{4} = 192$$

Cumulative frequency just greater than 192 is 219.

Hence $Q_3 = 5$.

$$\text{Here } \frac{4N}{10} = 4 \times 25.6 = 102.4$$

Cumulative frequency just greater than 102.4 is 167.

Hence $D_4 = 4$.

$$\text{Here } \frac{27N}{100} = 27 \times 2.56 = 69.12$$

Cumulative frequency just greater than 69.12 is 95. Hence $P_{27} = 3$.

For Continuous frequency distribution,

$$Q_1 = l + \frac{h}{f} \left(\frac{N}{4} - c \right)$$

$$Q_2 = l + \frac{h}{f} \left(\frac{N}{2} - c \right) = \text{median}$$

$$Q_3 = l + \frac{h}{f} \left(\frac{3N}{4} - c \right)$$

$$D_1 = l + \frac{h}{f} \left(\frac{N}{10} - c \right)$$

$$D_4 = l + \frac{h}{f} \left(\frac{4N}{10} - c \right)$$

$$D_7 = l + \frac{h}{f} \left(\frac{7N}{10} - c \right)$$

D_5 is the median.

$$P_9 = l + \frac{h}{f} \left(\frac{9N}{100} - c \right)$$

$$P_{27} = l + \frac{h}{f} \left(\frac{27N}{100} - c \right) \text{ etc.}$$

P_{50} is the median.

Find the median, lower and upper quartiles from the following table.

Marks :	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80
No. of students.	15	20	25	24	10	33	71	51

Solution:

Marks	No. of Student (f)	Cumulative frequency c.f.
0-10	15	15
10-20	20	35
20-30	25	60
30-40	24	84
40-50	10	94
50-60	33	127
60-70	71	198
70-80	51	249

Here $N = 249$

(i) Calculation of median:

$$\frac{N}{2} = \frac{249}{2} = 124.5$$

$\therefore$ median class is 50-60.

$$l=50, h=10, f=33, c=94$$

$$\begin{aligned}\text{Median} &= l + \frac{h}{f} \left(\frac{N}{2} - c \right) \\ &= 50 + \frac{10}{33} (124.5 - 94)\end{aligned}$$

$$\text{Median} = 59.24$$

(ii) Calculation of lower quartile (Q_1):

$$\frac{N}{4} = \frac{249}{4} = 62.25$$

$\therefore$ lower quartile class is 30-40

$$l=30, f=24, c=60, h=10$$

$$\therefore Q_1 = l + \frac{h}{f} \left(\frac{N}{4} - c \right)$$

$$Q_1 = 30 + \frac{10}{24} (62.25 - 60)$$

$$Q_1 = 30.94$$

(iii) Calculation of upper quartile (Q_3):

$$\frac{3N}{4} = \frac{747}{4} = 186.75$$

$\therefore$ upper quartile class is 60-70

$$l = 60, h = 10, f = 71, c = 127$$

$$Q_3 = l + \frac{h}{f} \left(\frac{3N}{4} - c \right)$$

$$= 60 + \frac{10}{71} (186.75 - 127)$$

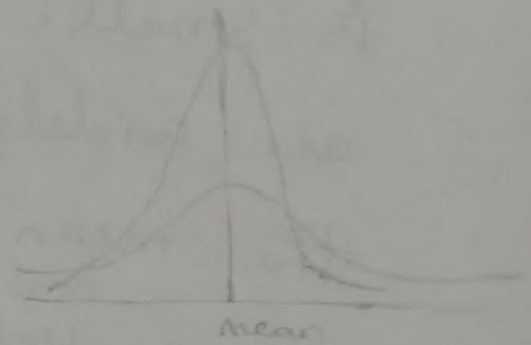
$$Q_3 = 68.42.$$

Dispersion (also called variability, scatter
or spread :

Although measures of central tendency do exhibit one of the important characteristics of a distribution yet they fail to give any idea as to how the individual values differ from the central value, i.e. whether they are closely packed around the central value or widely scattered away from it.

Two distributions may have the same mean and the same total frequency, yet they may differ in the extent to which the individual values may

be spread about the average (see fig)
The magnitude of such a variation is called dispersion.



Two (or) more frequency distributions may have exactly identical averages but even then they may differ markedly in several ways. Further analysis is, therefore, essential to account for these differences. Consider the following example.

Distribution A: 75 85 95 105 115 125
Distribution B: 10 20 30 70 180 290

A.M. of each distribution is $\frac{600}{6} = 100$.

In distribution A, the values of the variate differ from 100 but the difference is small. In distribution B, the items are widely scattered and lie far from the mean. Although the A.M. is the same, yet the two distributions widely differ from each other in their formation.

Therefore, while studying a distribution it is equally important to know how the variates are clustered around or scattered away from the point of central tendency. Such variation is called dispersion or spread or scatter or variability. Thus, dispersion is the extent to which the values are dispersed about the central value.

Ex:

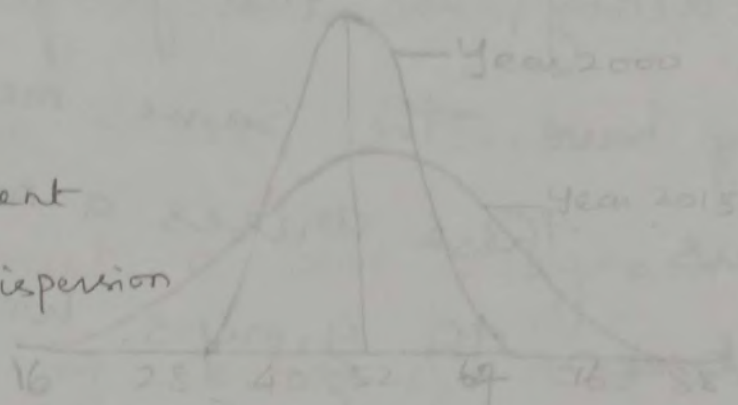
Centre : Same

Variation : Different

Year 2000: Close Dispersion

Year 2015: wide Dispersion

Better Quality Data: Data of Year 2000.



Ex:

The daily income of the workers in two factories are:

Factory A : 35 45 50 65 70 90 100

Factory B : 60 65 65 65 65 65 70

Here we observe that in both the groups the mean of the data is the same, namely, 65.

(i) In group A, the observations are much more scattered from mean.

(ii) In group B, almost all the observations are concentrated around the mean.

certainly, the two groups differ even though they have the same mean.

Thus, there arises a need to differentiate between the groups. We need some other measures which concern with the measure of scatteredness (or spread).

To do this, we study what is known as measure of dispersion.

Ex:

To explain the meaning of dispersion, let us consider an example.

Two sections of 10 students each in Class X in a certain school were given a common test in Mathematics (40 maximum marks). The scores of the students are given below.

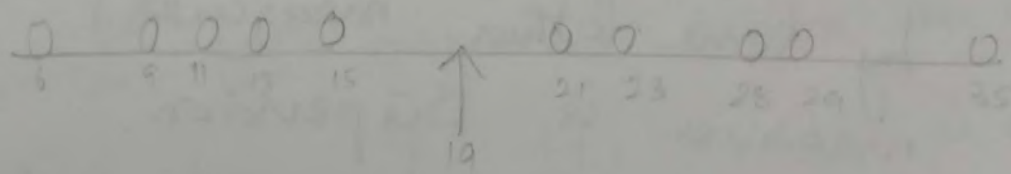
Section A: 6 9 11 13 15 21 23 28 29 35
Section B: 15 16 16 17 18 19 20 21 23 25

The average Score in Section A is 19.

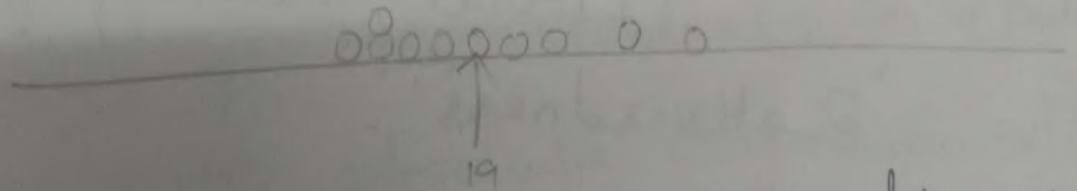
The average Score in Section B is 19.

Let us construct a dot diagram, on the same scale of Section A and Section B (see Figure). The position of mean is marked by an arrow in the dot diagram.

Section A:



Section B:



Clearly, the extent of spread or dispersion of the data is different in Section A from that of B. The measurement of the scatter of the given data about the average is said to be a measure of dispersion (or) scatter.

Dispersion:

Averages or the measures of central tendency gives us an idea of the concentration of the observations about the central part of the distribution. If we know the average alone we cannot form a complete idea about the distribution. The measures of central tendency are inadequate to give us a complete idea of the distribution. They must be supported and supplemented by some other measures. One such measure is Dispersion.

Literal meaning of dispersion is 'Scatteredness'...

Types of Dispersion:

The measures of dispersion can be either absolute or relative.

Absolute measures:

Absolute measures of dispersion are expressed in the same units in which the original data are expressed. For example, if the series is expressed as marks of the students in a particular Subject; the absolute dispersion will provide the value in marks. The only difficulty is that if two or more series are expressed in different units, the series cannot be compared on the basis of dispersion.

Relative (or) Coefficient Measures:

Relative (or) Coefficient of dispersion is the ratio or the percentage of a measure of absolute dispersion to an appropriate average. The basic advantage of this measure is that two or more series can be compared with each other despite the fact they are expressed in different units.

Theoretically, Absolute measure of dispersion is better. But from a practical point of view, relative or Coefficient of dispersion is considered better as it is used to make comparison between series.

Note:

For every absolute measure of dispersion, there is a relative measure.

Absolute Measures of Dispersion:

The measures of dispersion which are expressed in terms of original units of a data are termed as absolute measures.

Relative Measures of Dispersion:

Relative measures of dispersion are also known as Coefficient of dispersion. They are obtained as ratios or percentages. These are pure numbers independent of the units of measurement and used to compare two or more sets of data values.

The following are the measure of
dispersion:

Absolute Measures:

1. Range
2. Quartile Deviation (or) Semi-interquartile range
3. Mean Deviation
4. Standard Deviation.

Range:

Range is the simplest measure of dispersion and it is given by the difference between the maximum and the minimum observations in the data set.

It is denoted by $R = L - S$, where L is the largest value of variable and S is the smallest value of variable.

Ex:

If the weekly wages of a group of labourers are

Rs: 21 23 28 25 35 42 39 48

$$\begin{aligned}\text{then range} &= \text{Max. Value} - \text{Min. Value} \\ &= 48 - 21 = 27\end{aligned}$$

For grouped data,

Range = upper limit of last class - lower limit of first class.

Ex.

Discrete Series:

Marks of the
Students in Statistics
(out of 50) ×

No. of students

+

10

4

10

12

16

18

15

20

$$\text{Range} = 20 - 10 = 10 \text{ marks}$$

Continuous Series:

X

Frequencies

10-15

4

15-20

10

20-25

26

25-30

8

$$\text{Range} = 30 - 10 = 20 \text{ marks.}$$

Quantile deviation (or) Semi-interquartile range:

The difference between the upper and lower quantiles i.e. $Q_3 - Q_1$ is known as the inter quantile range and half of it i.e. $\frac{1}{2}(Q_3 - Q_1)$ is called the semi-inter quantile range or the quantile deviation.

Inter quantile range : $Q_3 - Q_1$

Semi-inter quantile range
(or) quantile deviation } $\frac{1}{2}(Q_3 - Q_1)$
(or) $\frac{(Q_3 - Q_2) + (Q_2 - Q_1)}{2}$

Q_1	Q_2	Q_3	
25%	25%	25%	25%

Interquantile
Range = $Q_3 - Q_1$

Merits of Range:

- * It is simple to understand and easy to calculate.
- * It is less time consuming.

Demerits of Range:

- * It is not based on all observations.
- * It gives a rough answer.
- * It is very much affected by the extreme values.
- * The value of Range is affected more by sampling fluctuations.
- * Range cannot be computed in case of open-end distribution.

Advantages of quantile deviation:

* It is a better measure of dispersion as it is not based on two extreme values like range but rather on middle 50% observations.

* It is the only measure of dispersion which can be used for open end distributions.

Disadvantages of quantile deviation:

* Quantile deviation is not based on complete items of the series as it ignores 50% of the observations.

* The value of Quartile deviation is much affected by sampling fluctuation.

* It requires some mathematical knowledge and time to compute quartile deviation. So, it is complex and time consuming method than range.

Mean Deviation (or) Average Deviation

For ungrouped data,

$$\text{Mean deviation from mean} = \frac{\sum_{i=1}^n |x_i - \bar{x}|}{N}$$

$$\text{where } N = \sum_{i=1}^n f_i$$

$$\text{Mean deviation from median} = \frac{\sum_{i=1}^n |x_i - \text{median}|}{N}$$

$$\text{Mean deviation from mode} = \frac{\sum_{i=1}^n |x_i - \text{mode}|}{N}$$

For grouped data,

$$\text{Mean deviation from mean} = \frac{\sum_{i=1}^n f_i |x_i - \bar{x}|}{N}$$

$$\text{Mean deviation from median} = \frac{\sum_{i=1}^n f_i |x_i - \text{median}|}{N}$$

$$\text{Mean deviation from mode} = \frac{\sum_{i=1}^n f_i |x_i - \text{mode}|}{N}$$

Merits of M.D

- * It is based on all the observations of a series.
- * It shown the dispersion or scatter of the various items of a series from its central value.
- * It is not very much affected by the values of extreme items of a series.
- * It facilitates comparison between different items of a series.
- * It truly represents the average of deviations of the items of a series.
- * It has practical usefulness in the field of business and commerce.

Demerits of M.D

* It is not rigidly defined in the sense that it is computed from any central value viz, mean, median, mode etc. and thereby it can produce different results.

* It violates the algebraic principle by ignoring the + and - signs while calculating the deviations of the different items from the central value of a series.

* It is not capable of further algebraic treatment.

* It is affected much by the fluctuation in sampling.

* It is not suitable for sociological study.

* It is difficult to calculate when the actual value of an average comes out in fraction, or recurring figure for that in such a case it requires the use of the shortcut method which involves a cumbersome formula subject to adjustment in different cases.

Standard deviation:

For ungrouped data,

$$S.D = \sigma = \sqrt{\frac{\sum (x_i - \bar{x})^2}{N}}$$

for grouped data,

$$S.D = \sigma = \sqrt{\frac{\sum f_i (x_i - \bar{x})^2}{N}}$$

$$(or) \sqrt{\frac{\sum f_i x_i^2}{N} - \left(\frac{\sum f_i x_i}{N}\right)^2}$$

Variance:

The square of S.D. is called the Variance.

$$ie \text{ Variance} = \sigma^2.$$

Root mean square deviation:

It is denoted by 's' and it is defined as $s = \sqrt{\frac{1}{N} \sum_i f_i (x_i - A)^2}$

where A is any arbitrary number.
's²' is called mean square deviation.

Short cut method for S.D. (change of origin)

$$\sigma = \sqrt{\frac{\sum fidi^2}{N} - \left(\frac{\sum fidi}{N}\right)^2}, \text{ where } d = x_i - A$$

Step-deviation method for S.D.

$$\sigma = h \sqrt{\frac{\sum f_i u_i^2}{N} - \left(\frac{\sum f_i u_i}{N}\right)^2} \quad u_i = \frac{x_i - A}{h}$$

(Shift of origin and change of scale.)

Merits of S.D

- * It is rigidly defined.
- * It is based on all the observations of the series.
- * It is amenable to further algebraic treatment.
- * It is least affected by fluctuations of sampling.

* The squaring of deviations makes them positive and the difficulty about algebraic signs which was experienced in case of mean deviation is not found here.

Demerits of S.D:

- * It is more affected by extreme items.
- * It cannot be exactly calculated for a distribution with open-ended classes.
- * Its calculation is difficult as it involves many mathematical models and processes.
- * It cannot be used for comparing the dispersion of two or more series given in different units.

Relation between measures of dispersion

$$M.D = \frac{4}{5} (S.D)$$

$$Q.D = \frac{2}{3} (S.D)$$

Relative measure of dispersion:

$$\text{Coefficient of range} = \frac{L - S}{L + S}$$

$$\text{Coefficient of Q.D} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

$$\text{Coefficient of M.D. from mean} =$$

$$\frac{\text{M.D from mean}}{\text{mean}}$$

$$\text{Coefficient of M.D. from median} =$$

$$\frac{\text{M.D. from median}}{\text{median}}$$

Coefficient of M.D. from mode =

$$(4.2) \frac{\text{M.D. from mode}}{\text{mode}}$$

$$\text{Coefficient of S.D} = \frac{\sigma}{\text{mean}}$$

$$\text{Coefficient of Variation} = 100 \times \frac{\sigma}{\text{mean}}$$

- ① Calculate: (i) Quartile deviation
(ii) mean deviation _{from mean} (iii) Standard deviation _{from mean}
for the following data.

Marks :	0-10	10-20	20-30	30-40	40-50	50-60	60-70
No. of Students :	6	5	8	15	7	6	3

Sol:

$$\text{mean } \bar{x} = A + h \frac{\sum fd}{N}$$

$$\text{M.D. from mean} = \frac{1}{N} \sum f |x - \bar{x}|$$

$$\text{S.D. from mean} = \sqrt{\frac{\sum f_i (x_i - \bar{x})^2}{N}}$$

$$\text{Q.D} = \frac{Q_3 - Q_1}{2}$$

Marks	Mid value (x)	No. of students (f)	c.f.	$d = \frac{x - 35}{10}$	fd	$ x - \bar{x} $	$f x - \bar{x} $
0-10	5	6	6	-3	-18	28.4	170.4
10-20	15	5	11	-2	-10	18.4	92
20-30	25	8	19	-1	-8	8.4	67.2
30-40	35	15	34	0	0	1.6	24
40-50	45	7	41	1	7	11.6	81.2
50-60	55	6	47	2	12	21.6	129.6
60-70	65	3	50	3	9	31.6	94.8
		<u>3</u>			<u>-8</u>		<u>659.2</u>
		<u>N=50</u>					

Here $N=50$, $\frac{N}{4} = 12.75$, $\frac{3}{4}N = 37.25$

Lower quartile class is 20-30.

$$Q_1 = l + \frac{h}{f} \left(\frac{N}{4} - c \right)$$

$$= 20 + \frac{10}{8} (12.75 - 11)$$

$$Q_1 = 22.19.$$

Upper quartile class is 40-50

$$Q_3 = l + \frac{h}{f} \left(\frac{3N}{4} - c \right)$$

$$= 40 + \frac{10}{7} (37.25 - 34)$$

$$Q_3 = 44.64$$

d^2	fd^2
9	54
4	20
1	8
0	0
1	7
4	24
9	27
	<u>140</u>

$$Q.D = \frac{Q_3 - Q_1}{2}$$

$$= \frac{44.64 - 22.19}{2}$$

$$Q.D = 11.23$$

$$\text{Mean} = \bar{x} = A + h \frac{\sum fd}{N}$$

$$= 35 + 10 \times \frac{(-8)}{50} = 33.4$$

$$S.D = \sqrt{\frac{\sum f_i(x_i - \bar{x})^2}{N}}$$

$$= \sqrt{\frac{13872}{50}}$$

$$= 16.66$$

$$x_i - \bar{x}$$

$$-28.4$$

$$-18.4$$

$$-8.4$$

$$1.6$$

$$11.6$$

$$21.6$$

$$31.6$$

$$(x_i - \bar{x})^2$$

$$806.56$$

$$338.56$$

$$70.56$$

$$2.56$$

$$134.56$$

$$466.56$$

$$998.56$$

$$M.D. \text{ from mean} = \frac{1}{N} \sum f|x - \bar{x}|$$

$$= \frac{659.2}{50} = 13.184$$

$$f_i(x_i - \bar{x})^2 \text{ S.D. } \sigma = h \sqrt{\frac{\sum f_i d_i^2}{N} - \left(\frac{\sum f_i d_i}{N}\right)^2}$$

$$4839.2$$

$$1692.8$$

$$564.64$$

$$38.4$$

$$941.92$$

$$2799.36$$

$$2995.84$$

$$13872$$

$$\text{where } d_i = \frac{x_i - A}{h}$$

$$= 10 \times \sqrt{\frac{140}{50} - \left(\frac{-8}{50}\right)^2}$$

$$= 10 \times \sqrt{2.8 - 0.0256}$$

$$= 10 \times \sqrt{2.7744} = 10 \times 1.66565$$

$$S.D = 16.66$$

Calculate the mean & Standard deviation for the following table giving the age distribution of 542 members.

Age in years:	20-30	30-40	40-50	50-60	60-70	70-80	80-90
No. of members:	3	61	132	153	140	51	2

Sol: $\bar{x} = 54.72$, Variance = $\sigma^2 = 141.07$

S.D = 11.88

Moment:

In mathematics, a moment is a specific quantitative measure of the shape of a function. It is used in both mechanics & statistics.

If the function represents physical density, then the zeroth moment is the total mass, the first moment divided by the total mass is the center of mass and the second moment is the rotational inertia.

If the function is a probability distribution, then the zeroth moment is the total probability (i.e. one), the first moment is the mean, the second central moment is the variance, the third standardized moment is the skewness, and the fourth standardized moment is the kurtosis.

for a distribution of mass or probability on a bounded interval, the collection of all the moments (of all orders from 0 to ∞) uniquely determines the distribution (Hausdorff moment problem). The same is not true on unbounded intervals (Hamburger moment problem).

The r^{th} moment about the mean $\bar{x}$ of a distribution is denoted by M_r and is given by

$$M_r = \frac{1}{N} \sum f_i (x_i - \bar{x})^r$$

The corresponding moment about any point 'a' is defined as

$$M'_r = \frac{1}{N} \sum f_i (x_i - a)^r$$

In particular,

$$\text{we have } \mu_0 = \mu_0' = 1$$

$$\mu_1 = 0, \mu_1' = \bar{x} - a, \mu_2 = \sigma^2$$

$$\mu_2 = \mu_2' - (\mu_1')^2$$

$$\mu_3 = \mu_3' - 3\mu_2'\mu_1' + 2\mu_1'^3$$

$$\mu_4 = \mu_4' - 4\mu_3'\mu_1' + 6\mu_2'\mu_1'^2 - 3\mu_1'^4$$

Skewness:

A symmetrical distribution, mean, median & mode are equal.

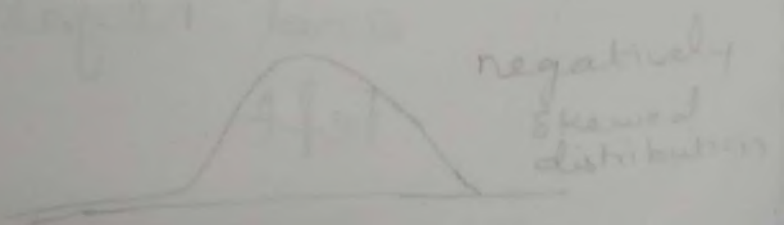
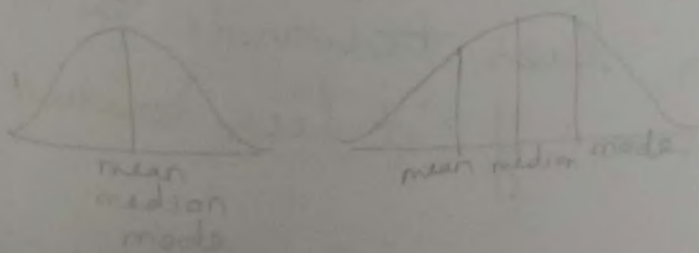
An asymmetrical distribution, mean, median & mode are not equal.

In a skewed distribution, values of mean, median & mode would not coincide.

Skewness Can be positive or negative or zero.

1. When the values of mean, median & mode are equal, there is no skewness.
2. When $\text{mean} > \text{median} > \text{mode}$, skewness will be positive.
3. When $\text{mean} < \text{median} < \text{mode}$, skewness will be negative.

When the skewness is presented in absolute term i.e. in units it is absolute skewness. If the value of skewness is obtained in ratios or percentages, it is called relative or coefficient of skewness.



Skewness means lack of symmetry or lopsidedness in a frequency distribution. The object of measuring skewness is to estimate the extent to which a distribution is distorted from a perfectly symmetrical distribution.

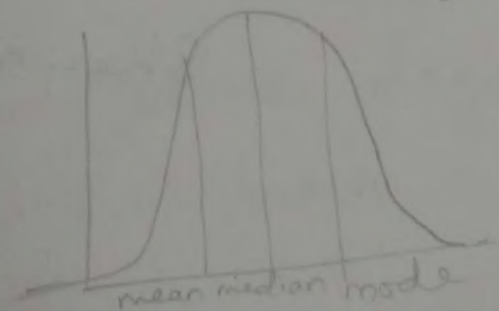
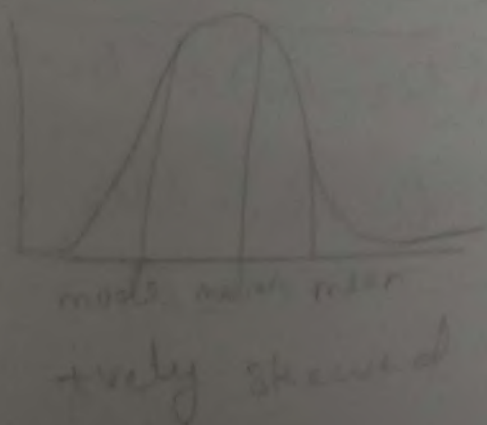
Skewness indicates whether the curve is turned more to one side than to other i.e. whether the curve has a longer tail on one side.

Skewness can be positive as well as negative. Skewness is positive if the longer tail of the distribution lies towards the right and negative if it lies towards the left.

Skewness measures the degree of asymmetry or the departure from symmetry.

If the frequency curve has a longer tail to the right i.e. the mean is to the right of the mode, then the distribution is said to have positive skewness.

If the curve is more elongated to the left, then it is said to have negative skewness.



1. Karl Pearson's Coefficient of Skewness is defined as

$$\text{Skewness} = \frac{\text{mean} - \text{mode}}{\sigma}$$

If the mode is ill-defined,

then use $\text{mode} = 3\text{median} - 2\text{mean}$

$$\therefore \text{Skewness} = \frac{3(\text{mean} - \text{median})}{\sigma}$$

The value of Karl Pearson's Coefficient of Skewness lies between -3 & $+3$.

2. Bowley's Coefficient of Skewness (based on quantiles) is defined as

$$\text{Skewness} = \frac{(Q_3 - Q_2) - (Q_2 - Q_1)}{(Q_3 - Q_2) + (Q_2 - Q_1)}$$

$$= \frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1}$$

$$= \frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1}$$

$$= \frac{Q_3 + Q_1 - 2Q_2}{Q_3 - Q_1}$$

The value of Bowley's coefficient of skewness lies between -1 & $+1$.

3. Karl Pearson Coefficient of Skewness (based on moments) is defined as

$$\gamma_1 = \sqrt{\beta_1} \quad \text{where} \quad \beta_1 = \frac{\mu_3^2}{\mu_2^3}$$

Thus $\gamma_1 = \sqrt{\beta_1}$ gives the simplest measure of skewness.

$\beta_1 = 0$ gives symmetric distribution

$\beta_1 > 0$ gives positively skewed distribution

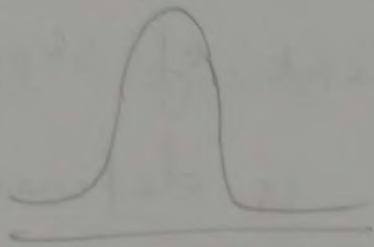
$\beta_1 < 0$ gives negatively skewed distribution

Kurtosis:

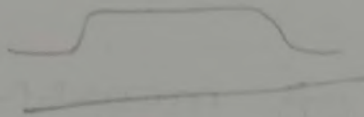
Kurtosis measures the degree of peakedness of a distribution and is

given by $\beta_2 = \frac{\mu_4}{\mu_2^2}$.

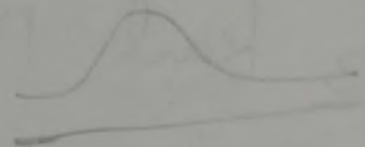
This measure of kurtosis is suggested by Karl Pearson.



Leptokurtic



platykurtic



mesokurtic

$\beta_2 = 3 \Rightarrow$ Curve is a mesokurtic curve or normal curve

$\beta_2 > 3 \Rightarrow$ Curve is a leptokurtic curve

$\beta_2 < 3 \Rightarrow$ Curve is a platykurtic curve.

Fisher introduced γ_2 coefficient as a measure of kurtosis. It is given by $\gamma_2 = \beta_2 - 3$

$\gamma_2 = 0 \Rightarrow$ Mesokurtic curve or normal curve

$\gamma_2 > 0 \Rightarrow$ Leptokurtic curve

$\gamma_2 < 0 \Rightarrow$ platykurtic curve

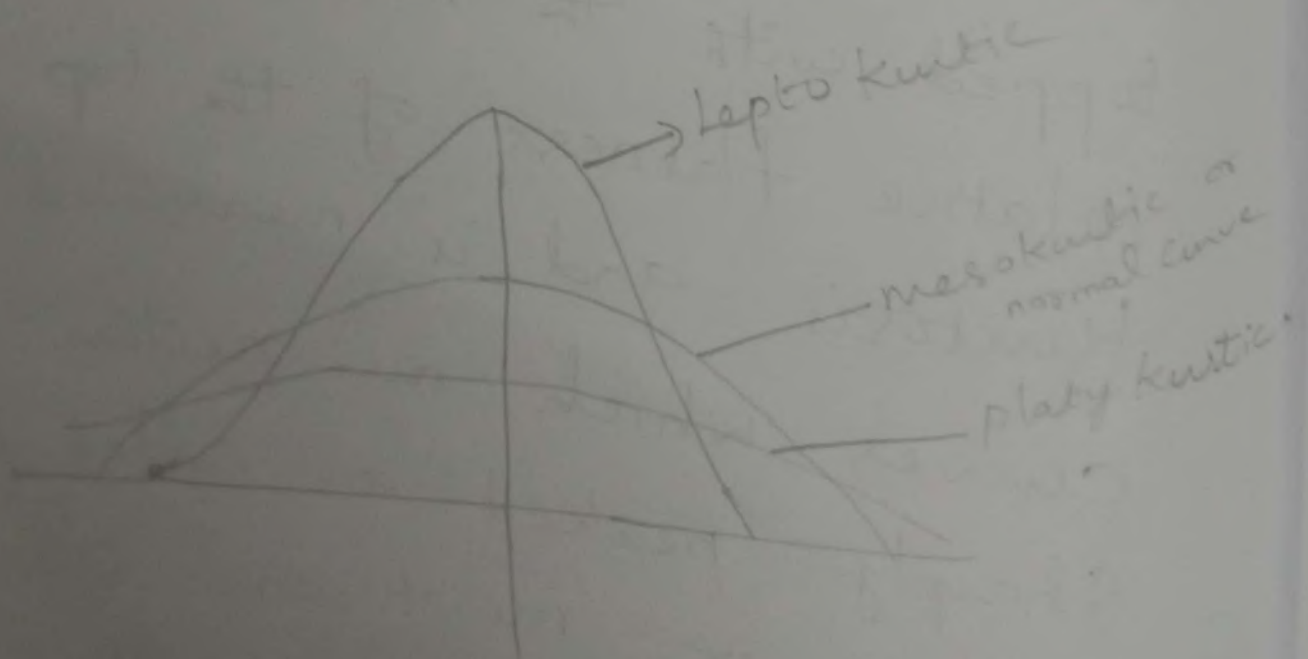
Kurtosis:

Given two frequency distributions which have the same variability as measured by the standard deviation, they may be relatively more or less flat topped than the 'normal curve'. A frequency curve may be symmetrical but it may not be equally flat topped with the normal curve. The relative flatness of the top is called kurtosis and is measured by β_2 . Curves which are neither flat nor sharply peaked are called normal curves or mesokurtic curves.

For such a curve $\beta_2 = 3$ and hence ν_2

Curves which are flatter than the normal curve are called platykurtic, for such a curve $\beta_2 < 3$ & hence $\nu_2 < 0$.

Curves which are more sharply peaked than the normal curve are called leptokurtic. For such a curve $\beta_2 > 3$ and hence $\nu_2 > 0$.



① From the following table showing the wage distribution in a certain factory, determine

- the mean wage
- The median wage
- The modal wage
- The wage limits for the middle 50% of the wage earners
- the percentages of workers who earned between Rs 75 and Rs 125.
- the Percentage who earned more than Rs 150 per week and
- the Percentage who earned less than Rs. 100 per week.

Ans: (a) $\bar{x} = 108.5$ (b) $med = 108.75$ (c) $Mo = 118.3$

(d) 81.25, 129.3 (e) 48 (f) 12 (g) 40.

weekly wages (Rs)	No. of employees	weekly wages	No. of employees
20-40	8	120-140	35
40-60	12	140-160	18
60-80	20	160-180	7
80-100	30	180-200	5
100-120	40		